

Chapter 1

Existence of a Universal Whittaker Vector for Twisted Rankin–Selberg Integrals

Abstract. This is our solution to Problem 2 from the First Proof benchmark [5]. We prove that for any generic irreducible admissible representation Π of $\mathrm{GL}_{n+1}(F)$ over a non-archimedean local field F , there exists a Whittaker function W in the Whittaker model of Π such that, for every generic irreducible admissible representation π of $\mathrm{GL}_n(F)$, a suitable choice of Whittaker function for π makes the twisted local Rankin–Selberg integral finite and nonzero for all $s \in \mathbb{C}$.

1.1 Problem

Let F be a non-archimedean local field with ring of integers $\mathfrak{o}$. Let N_r denote the subgroup of $\mathrm{GL}_r(F)$ consisting of upper-triangular unipotent elements. Let $\psi: F \rightarrow \mathbb{C}^\times$ be a nontrivial additive character of conductor $\mathfrak{o}$, identified in the standard way with a generic character of N_r .

Let Π be a generic irreducible admissible representation of $\mathrm{GL}_{n+1}(F)$, realized in its ψ^{-1} -Whittaker model $\mathcal{W}(\Pi, \psi^{-1})$. Must there exist $W \in \mathcal{W}(\Pi, \psi^{-1})$ with the following property?

Let π be a generic irreducible admissible representation of $\mathrm{GL}_n(F)$, realized in its ψ -Whittaker model $\mathcal{W}(\pi, \psi)$. Let $\mathfrak{q}$ denote the conductor ideal of π , let $Q \in F^\times$ be a generator of $\mathfrak{q}^{-1}$, and set

$$u_Q := I_{n+1} + Q E_{n,n+1} \in \mathrm{GL}_{n+1}(F),$$

where $E_{i,j}$ is the matrix with a 1 in the (i,j) -entry and 0 elsewhere. For some $V \in \mathcal{W}(\pi, \psi)$, the local Rankin–Selberg integral

$$\int_{N_n \backslash \mathrm{GL}_n(F)} W(\mathrm{diag}(g, 1)u_Q) V(g) |\det g|^{s-\frac{1}{2}} dg$$

is finite and nonzero for all $s \in \mathbb{C}$.

1.2 Introduction and statement of the result

1.2.1 Setup

Let F be a non-archimedean local field with ring of integers $\mathfrak{o}$, maximal ideal $\mathfrak{p}$, and uniformizer ϖ . Let q denote the cardinality of the residue field $\mathfrak{o}/\mathfrak{p}$.

For a positive integer m , write $G_m := \mathrm{GL}_m(F)$ for the general linear group, and $N_m \subset G_m$ for the subgroup of upper-triangular unipotent matrices. Fix a nontrivial additive character $\psi: F \rightarrow \mathbb{C}^\times$ of conductor

$\mathfrak{o}$ (that is, ψ is trivial on $\mathfrak{o}$ but nontrivial on $\mathfrak{p}^{-1}$). Identify ψ with a generic character of N_m via the standard convention

$$\psi(u) := \psi\left(\sum_{i=1}^{m-1} u_{i,i+1}\right), \quad u = (u_{ij}) \in N_m. \quad (1.1)$$

Let Π be a generic irreducible admissible representation of G_{n+1} , realized in its ψ^{-1} -Whittaker model $\mathcal{W}(\Pi, \psi^{-1})$. For $g \in G_{n+1}$, a function $W \in \mathcal{W}(\Pi, \psi^{-1})$ satisfies

$$W(ug) = \psi^{-1}(u) W(g) \quad \text{for all } u \in N_{n+1}.$$

Similarly, let π be a generic irreducible admissible representation of G_n , realized in its ψ -Whittaker model $\mathcal{W}(\pi, \psi)$. Let $\mathfrak{q}$ denote the conductor ideal of π , let $Q \in F^\times$ be a generator of $\mathfrak{q}^{-1}$, and define

$$u_Q := I_{n+1} + Q E_{n,n+1} \in G_{n+1},$$

where $E_{i,j}$ denotes the matrix with a 1 in the (i,j) -entry and 0 elsewhere.

1.2.2 The problem

The question asks whether there exists $W \in \mathcal{W}(\Pi, \psi^{-1})$ —depending only on Π —such that for every generic irreducible admissible π of G_n (with conductor data $\mathfrak{q}$, Q as above), there exists $V \in \mathcal{W}(\pi, \psi)$ for which the local Rankin–Selberg integral

$$\int_{N_n \backslash G_n} W(\text{diag}(g, 1) u_Q) V(g) |\det g|^{s-1/2} dg \quad (1.2)$$

is finite and nonzero for all $s \in \mathbb{C}$.

Theorem 1.2.1. *The answer is yes. For every $n \geq 1$ and every generic irreducible admissible representation Π of G_{n+1} , there exists $W \in \mathcal{W}(\Pi, \psi^{-1})$ such that for every generic irreducible admissible representation π of G_n , there exists $V \in \mathcal{W}(\pi, \psi)$ making the integral (1.2) a nonzero constant (independent of s), hence finite and nonzero for all $s \in \mathbb{C}$.*

The proof proceeds in three stages. In Section 1.3, we establish a matrix identity that converts the u_Q -twisted integral into a character-modified standard integral. In Section 1.4, we recall the Bernstein–Zelevinsky theory of mirabolic restriction and compact induction. In Section 1.5, we prove the theorem for $n \geq 2$ by constructing W and V with compact mirabolic support that forces exact cancellation in the integrand. In Section 1.6, we handle the case $n = 1$ separately using the classical nonvanishing of local Gauss sums.

1.3 The twisted value identity

Lemma 1.3.1. *For any $g \in G_n$ and $Q \in F^\times$, define $u_Q = I_{n+1} + Q E_{n,n+1}$. Then*

$$\text{diag}(g, 1) u_Q = \begin{pmatrix} I_n & Q g_{\bullet,n} \\ 0 & 1 \end{pmatrix} \text{diag}(g, 1), \quad (1.3)$$

where $g_{\bullet,n}$ denotes the n th column of g . Consequently, for any $W \in \mathcal{W}(\Pi, \psi^{-1})$,

$$W(\text{diag}(g, 1) u_Q) = \psi^{-1}(Q g_{nn}) W(\text{diag}(g, 1)). \quad (1.4)$$

Proof. Write $g = (g_{ij})_{1 \leq i,j \leq n}$. The product $\text{diag}(g, 1) u_Q$ equals

$$\begin{pmatrix} g & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} I_n & Q e_n \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} g & Q g e_n \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} g & Q g_{\bullet,n} \\ 0 & 1 \end{pmatrix},$$

where $e_n = (0, \dots, 0, 1)^T \in F^n$. On the other hand,

$$\begin{pmatrix} I_n & Q g_{\bullet,n} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} g & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} g & Q g_{\bullet,n} \\ 0 & 1 \end{pmatrix}.$$

This establishes (1.3).

The matrix $\begin{pmatrix} I_n & Q g_{\bullet,n} \\ 0 & 1 \end{pmatrix}$ lies in N_{n+1} , since it is upper-triangular unipotent in G_{n+1} . Its superdiagonal entries are $0, \dots, 0, Q g_{nn}$ (where $Q g_{nn}$ appears in position $(n, n+1)$), and all other above-diagonal entries contribute to positions $(i, n+1)$ with $i < n$, which do not appear in the generic character (1.1). Therefore, by the Whittaker equivariance of W ,

$$W\left(\begin{pmatrix} I_n & Q g_{\bullet,n} \\ 0 & 1 \end{pmatrix} \text{diag}(g, 1)\right) = \psi^{-1}(Q g_{nn}) W(\text{diag}(g, 1)). \quad \square$$

Remark 1.3.2. Identity (1.4) is purely a consequence of Whittaker equivariance. The left factor in (1.3) lies in the N_{n+1} -orbit direction; it produces a character multiplier but does not restrict the support of W on the quotient $N_{n+1} \backslash G_{n+1}$. In particular, even when $|Q|$ is large, no additional support cutoff is introduced.

By Lemma 1.3.1, the integral (1.2) equals

$$\Psi(s) := \int_{N_n \backslash G_n} \psi^{-1}(Q g_{nn}) W(\text{diag}(g, 1)) V(g) |\det g|^{s-1/2} dg. \quad (1.5)$$

1.4 Mirabolic restriction and compact induction

Definition 1.4.1. The *mirabolic subgroup* $P_m \subset G_m$ is defined by

$$P_m := \{g \in G_m : e_m^T g = e_m^T\},$$

where $e_m = (0, \dots, 0, 1)^T$. Equivalently, P_m consists of all matrices in G_m whose last row is $(0, \dots, 0, 1)$.

We have $N_m \subset P_m$, and P_m decomposes as a semidirect product: every element of P_m can be written uniquely as $\begin{pmatrix} h & v \\ 0 & 1 \end{pmatrix}$ with $h \in G_{m-1}$ and $v \in F^{m-1}$.

The following is a fundamental result of Bernstein and Zelevinsky [1, Theorem 5.21].

Proposition 1.4.2 (Bernstein–Zelevinsky [1, Theorem 5.21]). *Let σ be a generic irreducible admissible representation of G_m , and let ψ_σ denote the associated additive character (either ψ or ψ^{-1} , depending on the model). Then the restriction of the Whittaker model to the mirabolic subgroup contains the compactly induced representation:*

$$\mathcal{W}(\sigma, \psi_\sigma)|_{P_m} \supset \text{c-ind}_{N_m}^{P_m}(\psi_\sigma). \quad (1.6)$$

Here $\text{c-ind}_{N_m}^{P_m}(\psi_\sigma)$ denotes the space of locally constant functions $\Phi: P_m \rightarrow \mathbb{C}$ satisfying $\Phi(up) = \psi_\sigma(u) \Phi(p)$ for all $u \in N_m$ and $p \in P_m$, and having compact support modulo N_m .

The inclusion (1.6) means that for any such Φ , there exists a Whittaker function in $\mathcal{W}(\sigma, \psi_\sigma)$ whose restriction to P_m agrees with Φ . We apply this principle twice: once to Π on G_{n+1} (with character ψ^{-1}), and once to π on G_n (with character ψ).

Remark 1.4.3 (Hypothesis checks at each use of Proposition 1.4.2). At every invocation below, we verify the hypotheses locally:

1. At (1.7): $\sigma = \Pi$ on G_{n+1} , and Π is generic irreducible admissible by setup.
2. At (1.8): $\sigma = \pi$ on G_n , and π is generic irreducible admissible by setup.
3. At (1.10): $\sigma = \Pi$ on G_2 , with the same generic irreducible admissible hypothesis.

Thus the imported result is applied without changing hypotheses or conclusion.

1.5 Proof for $n \geq 2$

Fix $n \geq 2$ throughout this section.

1.5.1 Construction of W

Choose a compact open subgroup $J \subset P_n \cap G_n(\mathfrak{o})$. (For instance, one may take J to be any sufficiently small congruence subgroup of $G_n(\mathfrak{o})$ contained in P_n .) Since $J \subset G_n(\mathfrak{o})$, the character ψ is trivial on $N_n \cap J \subset N_n \cap G_n(\mathfrak{o})$, because ψ has conductor $\mathfrak{o}$ and the superdiagonal entries of elements of $N_n \cap G_n(\mathfrak{o})$ lie in $\mathfrak{o}$.

Definition 1.5.1. Define $\Phi: P_{n+1} \rightarrow \mathbb{C}$ by declaring:

- Φ is supported on $N_{n+1} \text{diag}(J, 1)$;
- for $u \in N_{n+1}$ and $j \in J$, $\Phi(u \text{diag}(j, 1)) := \psi^{-1}(u)$.

Lemma 1.5.2. *The function Φ is well-defined, locally constant, and lies in $\text{c-ind}_{N_{n+1}}^{P_{n+1}}(\psi^{-1})$.*

Proof. We must verify that the definition is consistent: if $u \text{diag}(j, 1) = u' \text{diag}(j', 1)$ with $u, u' \in N_{n+1}$ and $j, j' \in J$, then $\psi^{-1}(u) = \psi^{-1}(u')$.

From $u \text{diag}(j, 1) = u' \text{diag}(j', 1)$, we obtain $u'^{-1}u = \text{diag}(j'j^{-1}, 1)$. Since $j, j' \in J \subset P_n$ and J is a group, $j'j^{-1} \in J \subset P_n \cap G_n(\mathfrak{o})$. Now $\text{diag}(j'j^{-1}, 1) \in N_{n+1}$ if and only if $j'j^{-1} \in N_n$. In that case, $j'j^{-1} \in N_n \cap J \subset N_n \cap G_n(\mathfrak{o})$, so $\psi^{-1}(j'j^{-1}) = 1$ (since ψ has conductor $\mathfrak{o}$ and the superdiagonal entries lie in $\mathfrak{o}$). Thus $\psi^{-1}(u'^{-1}u) = \psi^{-1}(\text{diag}(j'j^{-1}, 1)) = 1$, giving $\psi^{-1}(u) = \psi^{-1}(u')$.

If $\text{diag}(j'j^{-1}, 1) \notin N_{n+1}$, then $u'^{-1}u$ is not in N_{n+1} , contradicting the fact that both u and u' are in N_{n+1} . So this case does not arise.

The function Φ satisfies $\Phi(up) = \psi^{-1}(u) \Phi(p)$ for $u \in N_{n+1}$ by construction, is locally constant because J is open, and has compact support modulo N_{n+1} (namely $\text{diag}(J, 1)$, which is compact). $\square$

By Proposition 1.4.2 (use 1 in Remark 1.4.3) applied to Π , choose $W \in \mathcal{W}(\Pi, \psi^{-1})$ with

$$W|_{P_{n+1}} = \Phi. \quad (1.7)$$

This W depends only on Π and the choice of J . It does not depend on π .

1.5.2 Support analysis of W on the embedding locus

Lemma 1.5.3. *With W as in (1.7), if $W(\text{diag}(g, 1)) \neq 0$ for some $g \in G_n$, then $g \in N_n J$. In particular, $g \in P_n$, and so $g_{nn} = 1$.*

Proof. Since $\text{diag}(g, 1) \in P_{n+1}$ (its last row is $(0, \dots, 0, 1) = e_{n+1}^T$), the restriction (1.7) gives $W(\text{diag}(g, 1)) = \Phi(\text{diag}(g, 1))$. By Definition 1.5.1, this is nonzero only if $\text{diag}(g, 1) \in N_{n+1} \text{diag}(J, 1)$.

Suppose $\text{diag}(g, 1) = u \text{diag}(j, 1)$ with $u \in N_{n+1}$ and $j \in J$. Writing u in block form as $u = \begin{pmatrix} u' & v \\ 0 & 1 \end{pmatrix}$ with $u' \in N_n$ and $v \in F^n$, we obtain

$$\begin{pmatrix} g & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} u' & v \\ 0 & 1 \end{pmatrix} \begin{pmatrix} j & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} u'j & v \\ 0 & 1 \end{pmatrix}.$$

Comparing entries: $v = 0$ and $g = u'j$. Since $u' \in N_n$ and $j \in J$, we conclude $g \in N_n J$.

Since $J \subset P_n$, every $j \in J$ has $j_{nn} = 1$, and every $u' \in N_n$ has $u'_{nn} = 1$. Therefore $g_{nn} = (u'j)_{nn} = 1$ (the (n, n) -entry of a product of matrices that are both 1 in that diagonal position, with u' upper-triangular unipotent). Explicitly, $g_{nn} = \sum_k u'_{nk} j_{kn} = u'_{nn} j_{nn} = 1$, where the last equality uses that $u'_{nk} = 0$ for $k < n$ (since $n \geq 2$ and u' is upper-triangular with 1s on the diagonal, the n th row of u' is e_n^T). $\square$

1.5.3 Construction of V and evaluation of the integral

Now fix an arbitrary generic irreducible admissible representation π of G_n , with conductor ideal $\mathfrak{q}$ and generator $Q \in F^\times$ of $\mathfrak{q}^{-1}$.

Definition 1.5.4. Define $\phi_Q: P_n \rightarrow \mathbb{C}$ by declaring:

- ϕ_Q is supported on $N_n J$;

- for $u \in N_n$ and $j \in J$, $\phi_Q(uj) := \psi(Q)\psi(u)$.

The same argument as in Lemma 1.5.2 (with ψ replacing ψ^{-1} and P_n replacing P_{n+1}) shows that ϕ_Q is well-defined and lies in $\text{c-ind}_{N_n}^{P_n}(\psi)$. By Proposition 1.4.2 (use 2 in Remark 1.4.3) applied to π , choose $V \in \mathcal{W}(\pi, \psi)$ with

$$V|_{P_n} = \phi_Q. \quad (1.8)$$

Proposition 1.5.5. *With W and V as constructed above, the integral (1.5) equals*

$$\Psi(s) = \text{vol}(N_n \backslash N_n J) > 0$$

for all $s \in \mathbb{C}$. In particular, it is finite and nonzero, independent of s .

Proof. By Lemma 1.5.3, the integrand in (1.5) is supported on the set of $g \in G_n$ with $g \in N_n J$. Since $N_n J \subset P_n \subset G_n$, the integral reduces to

$$\Psi(s) = \int_{N_n \backslash N_n J} \psi^{-1}(Q g_{nn}) W(\text{diag}(g, 1)) V(g) |\det g|^{s-1/2} dg. \quad (1.9)$$

We now evaluate each factor for $g \in N_n J$. Write $g = u'j$ with $u' \in N_n$ and $j \in J$.

Factor 1: g_{nn} . By Lemma 1.5.3, $g_{nn} = 1$, so $\psi^{-1}(Q g_{nn}) = \psi^{-1}(Q)$.

Factor 2: $W(\text{diag}(g, 1))$. Since $\text{diag}(g, 1) = \text{diag}(u', 1) \cdot \text{diag}(j, 1)$ with $\text{diag}(u', 1) \in N_{n+1}$, the Whittaker equivariance of Φ gives

$$W(\text{diag}(g, 1)) = \Phi(\text{diag}(u'j, 1)) = \psi^{-1}(u').$$

(Here we identify ψ^{-1} on N_{n+1} with ψ^{-1} on N_n via the embedding $N_n \hookrightarrow N_{n+1}$, $u' \mapsto \text{diag}(u', 1)$; the character (1.1) depends only on superdiagonal entries, which are the same for u' and $\text{diag}(u', 1)$.)

Factor 3: $V(g)$. By (1.8), $V(g) = \phi_Q(u'j) = \psi(Q)\psi(u')$.

Factor 4: $|\det g|^{s-1/2}$. Since $J \subset G_n(\mathfrak{o})$, we have $|\det j| = 1$ (the determinant of an element of $G_n(\mathfrak{o})$ is a unit in $\mathfrak{o}$, hence has absolute value 1). Since $\det u' = 1$ (as u' is unipotent), we obtain $|\det g| = |\det(u'j)| = 1$. Therefore $|\det g|^{s-1/2} = 1$ for all s .

Combining all factors. The integrand at $g = u'j$ equals

$$\psi^{-1}(Q) \cdot \psi^{-1}(u') \cdot \psi(Q) \cdot \psi(u') \cdot 1 = \psi^{-1}(Q) \psi(Q) \cdot \psi^{-1}(u') \psi(u') = 1 \cdot 1 = 1,$$

since $\psi^{-1}(x)\psi(x) = 1$ for all x .

Since the integrand is identically 1 on $N_n \backslash N_n J$ and vanishes elsewhere,

$$\Psi(s) = \int_{N_n \backslash N_n J} 1 dg = \text{vol}(N_n \backslash N_n J).$$

This volume is positive and finite because J is compact and open, so $N_n J$ is an open subset of P_n and $N_n \backslash N_n J \cong N_n \cap J \backslash J$ is compact with positive Haar measure. $\square$

Remark 1.5.6. The function W constructed above depends only on Π (and the auxiliary compact open J). The function V depends on π through the conductor-dependent constant $\psi(Q)$ in Definition 1.5.4, but the final integral is independent of π .

1.6 Proof for $n = 1$

When $n = 1$, we have $G_1 = F^\times$, $N_1 = \{1\}$, and every generic irreducible admissible representation π of G_1 is a quasicharacter $\chi: F^\times \rightarrow \mathbb{C}^\times$. We write $U := \mathfrak{o}^\times$ for the group of units.

1.6.1 Construction of W

The mirabolic subgroup of G_2 is

$$P_2 = \left\{ \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} : a \in F^\times, b \in F \right\}.$$

Definition 1.6.1. Define $\Phi: P_2 \rightarrow \mathbb{C}$ by

$$\Phi\left(\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}\right) := \mathbf{1}_U(a) \psi^{-1}(b),$$

where $\mathbf{1}_U$ denotes the characteristic function of U .

Lemma 1.6.2. $\Phi \in \text{c-ind}_{N_2}^{P_2}(\psi^{-1})$.

Proof. Here $N_2 = \{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} : b \in F \}$. For $u = \begin{pmatrix} 1 & c \\ 0 & 1 \end{pmatrix} \in N_2$ and $p = \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} \in P_2$, we have $up = \begin{pmatrix} a & b+c \\ 0 & 1 \end{pmatrix}$, so

$$\Phi(up) = \mathbf{1}_U(a) \psi^{-1}(b+c) = \psi^{-1}(c) \cdot \mathbf{1}_U(a) \psi^{-1}(b) = \psi^{-1}(u) \Phi(p).$$

The support of Φ modulo N_2 is $\{ \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix} : a \in U \}$, which is compact. $\square$

By Proposition 1.4.2 (use 3 in Remark 1.4.3) applied to Π on G_2 , choose $W \in \mathcal{W}(\Pi, \psi^{-1})$ with $W|_{P_2} = \Phi$. In particular,

$$W\left(\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}\right) = \mathbf{1}_U(a). \quad (1.10)$$

1.6.2 Evaluation of the integral

Fix a quasicharacter $\chi: F^\times \rightarrow \mathbb{C}^\times$ with conductor exponent $c = c(\chi) \geq 0$, meaning that χ is trivial on $1 + \mathfrak{p}^c$ (with c minimal) and the conductor ideal is $\mathfrak{q} = \mathfrak{p}^c$. Then $Q \in F^\times$ is a generator of $\mathfrak{q}^{-1} = \mathfrak{p}^{-c}$; we may take $Q = \varpi^{-c}$.

Set

$$V(a) := \chi(a), \quad a \in F^\times.$$

When $n = 1$, we have $N_1 = \{1\}$, so the Whittaker condition is vacuous and the Whittaker model of the generic representation $\pi = \chi$ is the 1-dimensional space $\{c\chi : c \in \mathbb{C}\}$. Hence $V \in \mathcal{W}(\chi, \psi)$.

No compact-support condition on V is needed here: compactness in the integral comes from W via (1.10), which restricts the domain to U .

By identity (1.4) (with $n = 1$, so $g = a \in F^\times$ and $g_{11} = a$), the integral (1.5) becomes

$$\Psi(s) = \int_{F^\times} \psi^{-1}(Qa) W\left(\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}\right) \chi(a) |a|^{s-1/2} d^\times a. \quad (1.11)$$

By (1.10), the integrand vanishes unless $a \in U$, and on U we have $V(a) = \chi(a)$, so

$$\Psi(s) = \int_U \psi^{-1}(Qa) \chi(a) d^\times a. \quad (1.12)$$

Since $|a| = 1$ for all $a \in U$, the factor $|a|^{s-1/2} = 1$, and $\Psi(s)$ is independent of s .

Proposition 1.6.3 (Primitive local Gauss sum nonvanishing (imported)). *Let $\chi: F^\times \rightarrow \mathbb{C}^\times$ be a quasicharacter of conductor exponent $c \geq 1$, and let $\psi': F \rightarrow \mathbb{C}^\times$ be an additive character with conductor ideal $\mathfrak{p}^c$. Then*

$$\mathfrak{g}(\chi, \psi') := \int_U \chi(a) \psi'(a) d^\times a$$

is nonzero.

Proof. This is the standard primitive local Gauss-sum formula in Tate's local theory of epsilon factors [4, Section 3]; see also the local constant computation in [3, Section 2]. For matching conductors, the above integral is a nonzero scalar multiple of $\varepsilon(1/2, \chi, \psi')$, and local epsilon factors are nonzero. $\square$

Proposition 1.6.4. *The integral (1.12) is nonzero for every quasicharacter χ of $F^\times$.*

Proof. We consider two cases based on the conductor exponent $c = c(\chi)$.

Case 1: $c = 0$ (unramified χ). Then $\chi|_U \equiv 1$ and $Q \in \mathfrak{o}^\times$ (since $\mathfrak{q}^{-1} = \mathfrak{o}$). For $a \in U = \mathfrak{o}^\times$, we have $Qa \in \mathfrak{o}$, so $\psi^{-1}(Qa) = 1$ (since ψ has conductor $\mathfrak{o}$, meaning ψ is trivial on $\mathfrak{o}$). Therefore

$$\Psi(s) = \int_U 1 d^\times a = \text{vol}(U) \neq 0.$$

Case 2: $c \geq 1$ (ramified χ). Here $Q = \varpi^{-c}$ (up to a unit, which we may absorb). The additive character $a \mapsto \psi^{-1}(Qa) = \psi^{-1}(\varpi^{-c}a)$ has conductor ideal $\mathfrak{p}^c$: it is trivial on $\mathfrak{p}^c$ but nontrivial on $\mathfrak{p}^{c-1}$. The multiplicative character $\chi|_U$ has conductor exponent c : it is trivial on $1 + \mathfrak{p}^c$ but nontrivial on $1 + \mathfrak{p}^{c-1}$.

Therefore (1.12) is a primitive local Gauss sum:

$$\Psi(s) := \int_U \psi^{-1}(Qa) \chi(a) d^\times a$$

with matching conductor exponent c . Applying Proposition 1.6.3 with $\psi' = \psi^{-1}(Q \cdot)$ gives $\Psi(s) \neq 0$. $\square$

Remark 1.6.5. The key point in Case 2 is that the conductors match: the additive character ψ_Q restricted to U has conductor c (it is trivial on $1 + \mathfrak{p}^c$ since $\psi^{-1}(Q(1+x)) = \psi^{-1}(Q)\psi^{-1}(Qx)$ and $Qx \in \mathfrak{p}^c \cdot \mathfrak{p}^{-c} = \mathfrak{o}$ for $x \in \mathfrak{p}^c$), and the multiplicative character χ also has conductor c . A Gauss sum with mismatched conductors can vanish, but a primitive one (matching conductors) never does.

1.7 Conclusion

Combining Sections 1.5 and 1.6, we have established Theorem 1.2.1: for every $n \geq 1$ and every generic irreducible admissible representation Π of $\text{GL}_{n+1}(F)$, there exists $W \in \mathcal{W}(\Pi, \psi^{-1})$ such that for every generic irreducible admissible π of $\text{GL}_n(F)$, there exists $V \in \mathcal{W}(\pi, \psi)$ making the twisted Rankin–Selberg integral (1.2) finite and nonzero for all $s \in \mathbb{C}$.

In both cases ($n \geq 2$ and $n = 1$), the integral evaluates to a nonzero constant independent of s . The construction of W uses the Bernstein–Zelevinsky compact-induction embedding to prescribe $W|_{P_{n+1}}$ with mirabolic support forcing, while V is chosen (depending on π) to produce exact cancellation of all character factors in the integrand.

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